Similarity Index	Formula	Invariant to: Invertible Linear Transform	Orthogonal Transform	Isotropic Scaling
Linear Reg. (R^2_{LR})	$\|\|Q_Y^{\text{T}}X\|\|_{\text{F}}^2/\|\|X\|\|_{\text{F}}^2$	Y only	✓	✓
CCA (R^2_{CCA})	$\|\|Q_Y^{\text{T}}Q_X\|\|_{\text{F}}^2/p_1$	✓	✓	✓
CCA ($\bar{\rho}_{\text{CCA}}$)	$\|\|Q_Y^{\text{T}}Q_X\|\|_*/p_1$	✓	✓	✓
SVCCA (R^2_{SVCCA})	$\|\|(U_YT_Y)^{\text{T}}U_XT_X\|\|_{\text{F}}^2/\min(\|\|T_X\|\|_{\text{F}}^2, \|\|T_Y\|\|_{\text{F}}^2)$	If same subspace kept	✓	✓
SVCCA ($\bar{\rho}_{\text{SVCCA}}$)	$\|\|(U_YT_Y)^{\text{T}}U_XT_X\|\|_*/\min(\|\|T_X\|\|_{\text{F}}^2, \|\|T_Y\|\|_{\text{F}}^2)$	If same subspace kept	✓	✓
PWCCA	$\sum_{i=1}^{p_1} \alpha_i\rho_i/\|\|\alpha\|\|_1$, $\alpha_i = \sum_j \|\langle \mathbf{h}_i, \mathbf{x}_j\rangle\|$	✗	✗	✓
Linear HSIC	$\|\|Y^{\text{T}}X\|\|_{\text{F}}^2/(n-1)^2$	✗	✓	✗
Linear CKA	$\|\|Y^{\text{T}}X\|\|_{\text{F}}^2/(\|\|X^{\text{T}}X\|\|_{\text{F}}\|\|Y^{\text{T}}Y\|\|_{\text{F}})$	✗	✓	✓
RBF CKA	$\text{tr}(KHLH)/\sqrt{\text{tr}(KHKH)\text{tr}(LHLH)}$	✗	✓	✓*

Table 1. Summary of similarity methods investigated. Q_X and Q_Y are orthonormal bases for the columns of X and Y. U_X and U_Y are the left-singular vectors of X and Y sorted in descending order according to the corresponding singular vectors. $||\cdot||_*$ denotes the nuclear norm. T_X and T_Y are truncated identity matrices that select left-singular vectors such that the cumulative variance explained reaches some threshold. For RBF CKA, K and L are kernel matrices constructed by evaluating the RBF kernel between the examples as in Section 3, and H is the centering matrix $H_n = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^{\text{T}}$. See Appendix C for more detail about each technique.
*Invariance of RBF CKA to isotropic scaling depends on the procedure used to select the RBF kernel bandwidth parameter. In our experiments, we selected the bandwidth as a fraction of the median distance, which ensures that the similarity index is invariant to isotropic scaling.

For a linear kernel, CKA is equivalent to the RV coefficient (Robert & Escoufier, 1976) and to Tucker's congruence coefficient (Tucker, 1951; Lorenzo-Seva & Ten Berge, 2006).

Kernel Selection. Below, we report results of CKA with a linear kernel and the RBF kernel $k(\mathbf{x}_i, \mathbf{x}_j) = \exp(-||\mathbf{x}_i - \mathbf{x}_j||_2^2/(2\sigma^2))$. For the RBF kernel, there are several possible strategies for selecting the bandwidth σ, which controls the extent to which similarity of small distances is emphasized over large distances. We set σ as a fraction of the median distance between examples. In practice, we find that RBF and linear kernels give similar results across most experiments, so we use linear CKA unless otherwise specified. Our framework extends to any valid kernel, including kernels equivalent to neural networks (Lee et al., 2018; Jacot et al., 2018; Garriga-Alonso et al., 2019; Novak et al., 2019).

4. Related Similarity Indexes

In this section, we briefly review linear regression, canonical correlation, and other related methods in the context of measuring similarity between neural network representations. We let Q_X and Q_Y represent any orthonormal bases for the columns of X and Y, *i.e.* $Q_X = X(X^{\text{T}}X)^{-1/2}$, $Q_Y = Y(Y^{\text{T}}Y)^{-1/2}$ or orthogonal transformations thereof. Table 1 summarizes the formulae and invariance properties of the indexes used in experiments. For a comprehensive general review of linear indexes for measuring multivariate similarity, see Ramsay et al. (1984).

Linear Regression. A simple way to relate neural network representations is via linear regression. One can fit every feature in X as a linear combination of features from Y. A suitable summary statistic is the total fraction of variance explained by the fit:

$$R^2_{\text{LR}} = 1 - \frac{\min_B ||X - YB||_{\text{F}}^2}{||X||_{\text{F}}^2} = \frac{||Q_Y^{\text{T}}X||_{\text{F}}^2}{||X||_{\text{F}}^2}. \quad (5)$$

We are unaware of any application of linear regression to measuring similarity of neural network representations, although Romero et al. (2015) used a least squares loss between activations of two networks to encourage thin and deep "student" networks to learn functions similar to wide and shallow "teacher" networks.

Canonical Correlation Analysis (CCA). Canonical correlation finds bases for two matrices such that, when the original matrices are projected onto these bases, the correlation is maximized. For $1 \leq i \leq p_1$, the ith canonical correlation coefficient ρ_i is given by:

$$\begin{aligned} \rho_i = \max_{\mathbf{w}_X^i, \mathbf{w}_Y^i} & \text{corr}(X\mathbf{w}_X^i, Y\mathbf{w}_Y^i) \\ \text{subject to} \quad & \forall_{j<i} \; X\mathbf{w}_X^i \perp X\mathbf{w}_X^j \\ & \forall_{j<i} \; Y\mathbf{w}_Y^i \perp Y\mathbf{w}_Y^j. \end{aligned} \quad (6)$$

The vectors $\mathbf{w}_X^i \in \mathbb{R}^{p_1}$ and $\mathbf{w}_Y^i \in \mathbb{R}^{p_2}$ that maximize ρ_i are the canonical weights, which transform the original data into canonical variables $X\mathbf{w}_X^i$ and $Y\mathbf{w}_Y^i$. The constraints in (6) enforce orthogonality of the canonical variables.

For the purpose of this work, we consider two summary